

Gut metagenomics-derived genes as potential biomarkers of Parkinson's disease

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Identification of the gut microbiome compositions associated with disease has become a research focus worldwide. Emerging evidence has revealed the presence of gut microbiota dysbiosis in Parkinson's disease. In this study, we aimed to identify the gut microbiome associated with Parkinson's disease and subsequently to screen and to validate potential diagnostic biomarkers of Parkinson's disease. This case-control study investigated gut microbial genes in faeces from 40 volunteer Chinese patients with Parkinson's disease and their healthy spouses using shotgun metagenomic sequencing. Furthermore, the identified specific gut microbial gene markers were validated with real-time PCR in an independent Chinese cohort of 78 Parkinson's disease patients, 75 control subjects, 40 patients with multiple system atrophy and 25 patients with Alzheimer's disease. We developed the first gut microbial gene catalogue associated with Parkinson's disease. Twenty-five gene markers were identified that distinguished Parkinson's disease patients from healthy control subjects, achieving an area under the receiver operating characteristic curve (AUC) of 0.896 (95% confidence interval: 83.1–96.1%). A highly accurate Parkinson's disease index, which was not influenced by disease severity or Parkinson's disease medications, was created. Testing these gene markers using quantitative PCR distinguished Parkinson's disease patients from healthy controls not only in the 40 couples (AUC = 0.922, 95% confidence interval: 86.4–98.0%), but also in an independent group of 78 patients with Parkinson's disease and 75 healthy control subjects (AUC = 0.905, 95% confidence interval: 86.0–95.1%). This classifier also performed a differential diagnosis power in discriminating these 78 patients with Parkinson's disease from a cohort of 40 patients with multiple system atrophy and 25 patients with Alzheimer's disease based on the panel of 25 biomarkers. Based on our results, the identified Parkinson's disease index based on the gene set from the gut microbiome may be a potential diagnostic biomarker of Parkinson's disease.

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Abbreviations: AUC = area under the receiver operating characteristic curve; MGS = metagenomic species; MSA = multiple system atrophy; MWAS = metagenome-wide association study; PDI = Parkinson's disease index

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Introduction

Parkinson's disease is a chronic and progressive neurodegenerative disease characterized by the loss of dopaminergic neurons and the formation of Lewy pathology [mainly α -synuclein (α -syn)] in the substantia nigra. Parkinson's disease comprises motor and non-motor symptoms and affects 1% of the population aged ≥ 60 years worldwide (de Lau and Breteler, 2006) and 1.7% of the population aged ≥ 65 years in China (Zhang *et al.*, 2005). A wide range of measurements have been evaluated as biomarkers of idiopathic Parkinson's disease and mainly include clinical symptoms, tissue and fluid tests and functional neuroimaging data (Cersosimo *et al.*, 2013; Adams-Carr *et al.*, 2016). The reliability and reproducibility of the tissue and fluid tests have been questioned. Moreover, functional neuroimaging tests are quite expensive and can be performed at only a small number of centres. To date, the diagnosis of Parkinson's disease largely depends on clinical manifestations and patients' medical history. Therefore, identification of reliable and cost-effective biomarkers of Parkinson's disease is urgently needed. Constipation, one of the most common non-motor symptoms, appears throughout all stages of Parkinson's disease, even before the onset of motor symptoms, and increases with disease progression (Braak *et al.*, 2006; Shannon *et al.*, 2012b). According to the Movement Disorders Society (MDS) research criteria for prodromal Parkinson's disease, constipation is considered a clinical biomarker (Berg *et al.*, 2015). Furthermore, the formation of α -syn in the colon is detected earlier than the appearance of Parkinson's disease-related motor symptoms in both patients with Parkinson's disease (Cersosimo and Benarroch, 2012; Shannon *et al.*, 2012a) and Parkinson's disease mouse models (Braak *et al.*, 2006; Bencsik *et al.*, 2014). The Braak hypothesis proposes that α -syn first aggregates in the enteric nervous system and is then transported to the CNS (Braak *et al.*, 2006; Bencsik *et al.*, 2014). However, the detection of α -syn in the colon mucosa requires a colonoscopy to obtain a biopsy, limiting its broad application. Recently, the gut microbiota has been suggested to play an important role in the progression (especially with regard to α -syn aggregation) of Parkinson's disease in mouse models through the microbiota-gut-brain axis (Sampson *et al.*, 2016; Yang *et al.*, 2018). Is the gut microbiota a reliable biomarker of Parkinson's disease?

Gut microbiota dysbiosis has been observed in Parkinson's disease patients worldwide, and many studies have identified specific taxa associated with Parkinson's disease based on 16S rRNA gene amplicon sequencing (Keshavarzian *et al.*, 2015; Scheperjans *et al.*, 2015; Hill-Burns *et al.*, 2017; Hopfner *et al.*, 2017; Li *et al.*, 2017b, 2019b; Petrov *et al.*, 2017; Heintz-Buschart *et al.*, 2018; Lin *et al.*, 2018, 2019; Qian *et al.*, 2018; Aho *et al.*, 2019; Barichella *et al.*, 2019; Pietrucci *et al.*, 2019) or quantitative PCR (Hasegawa *et al.*, 2015; Unger *et al.*, 2016; Minato *et al.*, 2017), suggesting that the gut microbiota represents a potential biomarker of Parkinson's disease. Some

studies have proposed diagnostic models of Parkinson's disease based on taxonomic information, but the specific taxa were different among the studies (Scheperjans *et al.*, 2015; Bedarf *et al.*, 2017; Lin *et al.*, 2018; Qian *et al.*, 2018). Compared with the 16S rRNA gene amplicon method, shotgun metagenomic sequencing has multiple advantages, including an increased predictive ability of genes, enhanced detection of microbial species and detailed descriptions of the functions of the identified genes, enabling acquisition of genetic information regarding the entire community of microorganisms (Jovel *et al.*, 2016; Ranjan *et al.*, 2016). Shotgun metagenomic sequencing has been used to survey the associations between the gut microbiome and various disorders, such as type 2 diabetes (Qin *et al.*, 2012; Karlsson *et al.*, 2013), liver cirrhosis (Qin *et al.*, 2014), colorectal cancer (Petrov *et al.*, 2017), atherosclerosis (Karlsson *et al.*, 2012), and ankylosing spondylitis (Wen *et al.*, 2017). Furthermore, metagenome-wide association studies (MWASs) have been used to detect microbiome-based gene markers for the diagnosis of diseases (Wang and Jia, 2016), e.g. liver cirrhosis (Qin *et al.*, 2014), type 2 diabetes (Qin *et al.*, 2012), and colorectal cancer (Petrov *et al.*, 2017). To date, only one metagenomic analysis of Parkinson's disease has been reported, but it did not present a detailed genetic analysis (Bedarf *et al.*, 2017). However, detailed information about gut microbial genes is needed to explore their impacts on Parkinson's disease. In particular, given the costly and time-consuming nature of metagenomic sequencing, an economic and easy-to-use method of validating selected gene markers is required for use in clinical practice.

Here, we performed a case-control study based on shotgun metagenomic sequencing of DNA extracted from faecal samples from Parkinson's disease patients and their healthy spouses in a Chinese Han population using MWASs to identify biomarkers of Parkinson's disease. Furthermore, we developed a fast and sensitive real-time PCR method for verifying Parkinson's disease-specific gut microbiota gene markers identified using metagenomic sequencing and for validating the data in an independent cohort of Parkinson's disease patients, multiple system atrophy (MSA) patients and Alzheimer's disease patients.

Materials and methods

Study design and patient recruitment

Each participant was informed of the purpose of the study, and all enrolled subjects provided written informed consent. This study protocol was approved by the Research Ethics Committee, Ruijin Hospital, Shanghai Jiaotong University School of Medicine, Shanghai, China.

All Parkinson's disease patients eligible for this study were diagnosed with idiopathic Parkinson's disease according to the UK Brain Bank criteria (Daniel and Lees, 1993). The patient exclusion criteria were as follows: (i) atypical or secondary

parkinsonism; (ii) serious illness (e.g. heart failure or malignancy); (iii) inflammatory gastrointestinal disease; (iv) chronic disease that might influence the gut microbiota (e.g. diabetes, liver cirrhosis or cardiovascular disease); (v) haematological or autoimmune disease or the use of immunosuppressive agents in the past 3 months; and (vi) antibiotic use within 3 months prior to sample collection. The healthy controls exhibited no disease symptoms and did not satisfy the exclusion criteria.

For metagenomic sequencing, 66 idiopathic Chinese Han Parkinson's disease patients and their healthy spouses living in the same household were recruited from the Movement Disorders Clinic at the Department of Neurology of Ruijin Hospital. However, some participants were unwilling to provide samples; thus, 59 sample pairs were sent for quality testing to Shanghai Biotechnology Corporation. From the quality testing and matching, 40 pairs met the requirements for building a shotgun sequencing database. Forty patients [21 (52.5%) female; mean (standard deviation, SD) age 66.6 (7.1) years] and their healthy spouses [19 (47.5%) female, age 66.3 (8.1) years] were included in the final analysis (see the recruitment flowchart in [Supplementary Fig. 1](#)). Each couple included in this study had lived in the same household for at least 20 years.

An independent group of 78 Parkinson's disease patients [37 (47.4%) female; age 67.0 (5.6) years], 75 healthy control subjects [36 (48.0%) female; age 65.3 (7.5) years], 40 patients with MSA [17 (42.5%) female; age 61.0 (6.7) years] and 25 patients with Alzheimer's disease [13 (52.0%) female; age 66.1 (5.2) years] was enrolled as a test cohort for real-time PCR analysis to validate the discriminatory power of our method. All patients with MSA were categorized as probable or possible MSA based on consensus criteria ([Gilman *et al.*, 2008](#)). The patients with MSA had an average disease duration of 3.9 (1.9) years and were classified by the clinical phenotype of MSA parkinsonism-dominant subtype (MSA-P) ($n = 23$) or MSA cerebellar ataxia-dominant subtype (MSA-C) ($n = 17$) based on the predominant symptom complex and examination findings. All patients with Alzheimer's disease were defined according to the National Institute on Aging and Alzheimer's Association (NIA-AA) diagnostic guidelines for dementia due to probable Alzheimer's disease ([McKhann *et al.*, 2011](#)).

Clinical data collection

Clinical data were collected through in-person interviews with movement disorder specialists. Each subject's weight and height were measured and then used to calculate body mass index (BMI). Lifestyle habits of cigarette, alcohol, tea, coffee and yoghurt consumption were also recorded for all individuals. Participation in a habit was defined as participation at least once a day for the last 3 months. All the individuals involved in our study were omnivorous, and no individuals took probiotic supplements. Clinical characteristics included the age at onset, disease duration, Hoehn and Yahr stage, motor symptoms/non-motor symptoms and medication use. The anti-parkinsonism medications included levodopa, dopamine agonists (pramipexole/piribedil), monoamine oxidase B (MAO-B) inhibitor (selegiline), catechol-O-methyltransferase (COMT) inhibitor (entacapone), benzhexol hydrochloride and amantadine. The motor complications were diagnosed according to the Unified Parkinson's Disease Rating Scale (UPDRS) Part IV(A) and IV(B)

([Chapuis *et al.*, 2005](#)). The patients with MSA were assessed with the Unified Multiple System Atrophy Rating Scale (UMSARS). The UPDRS scores and Hoehn and Yahr stages of the patients were examined during the ON state. Levodopa equivalent doses (LEDs) were calculated using a previously described method ([Tomlinson *et al.*, 2010](#)). Non-motor symptoms were evaluated using the Non-Motor Symptoms Questionnaire for Parkinson's disease (NMS-Quest), Hamilton Anxiety Scale (HAMA), Hamilton Depression Scale (HAMD) and Mini-Mental State Examination (MMSE). Constipation was assessed using the Rome III criteria.

Sample collection and DNA extraction

Each study participant was asked to collect a faecal sample in the morning using a faecal collection container. The containers were transferred on ice and stored at -80°C prior to processing. Total faecal DNA was extracted using a QIAamp DNA Stool Mini Kit (Qiagen) according to the manufacturer's instructions. All DNA extraction procedures were performed in a Class II biologic safety cabinet. The concentration of genomic DNA in each sample was quantified using a NanoDrop 2000 spectrophotometer (Thermo Scientific).

Shotgun sequencing

The extracted microbial DNA was processed to construct metagenome shotgun sequencing libraries according to the manufacturer's instructions (Illumina). Each library was sequenced with the Illumina HiSeq X-ten platform using a PE150 strategy at Shanghai Biotechnology Corporation. We used the same workflow as described elsewhere to perform cluster generation, template hybridization, isothermal amplification, linearization, blocking and denaturation, and hybridization of the sequencing primers ([Supplementary material](#)).

Associations between profiles and clinical characteristics

The effects of the covariates on all profiles were assessed by permutational multivariate ANOVA (PERMANOVA) using the method implemented in the R package 'vegan'. The permuted P -value was obtained using 999 permutations ([Supplementary material](#)).

Gut microbial gene markers for identifying Parkinson's disease

Metagenomic species (MGS) were clustered as described in a previous study ([Nielsen *et al.*, 2014](#)) and based on genes that differed in abundance between the patients with Parkinson's disease and healthy control subjects ($P < 0.05$, Wilcoxon rank sum test). Fifteen MGS containing 51 816 genes were selected based on the following criteria: (i) the number of genes contained in the MGS was > 50 ; (ii) the MGS was assigned to the genus to which it aligns in the genome; and (iii) the ratio of annotation to the genome was at least 90%. The gut microbiota gene markers used to classify the Parkinson's disease patients

were selected using the minimum redundancy–maximum relevance (mRMR) method implemented in the R package ‘sideChannelAttack’ (Ding and Peng, 2005; Peng *et al.*, 2005). A set of 25 microbial genes was selected as the optimal classification set to identify Parkinson’s disease. On the basis of these 25 gene markers, a support vector machine (SVM) classifier (linear function kernel and default parameters) was constructed for patient discrimination in the R package ‘e1071’, the performance of which was assessed by receiver operating characteristic (ROC) analysis. The area under the ROC curve (AUC) and corresponding 95% confidence intervals (CIs) were obtained using the R package ‘pROC’ (10 000 bootstrap replicates).

A more straightforward index for patient discrimination, the Parkinson’s disease index (PDI), was defined to facilitate the clinical application of the selected microbial gene markers. Calculation of the PDI is shown in the [Supplementary material](#).

Detection of the selected gene markers using real-time PCR

The levels of the 25 genes were tested in the 80 individuals analysed using the shotgun sequencing approach as a training dataset. Another larger group including 78 Parkinson’s disease patients, 75 healthy control subjects and 40 MSA patients was used for testing data validation. Plasmids containing the 25 gene markers were constructed for use as DNA standards in the following real-time PCR experiment. Sample amplification and standard dilutions were performed in triplicate on an ABI ViiA7 instrument detection system (Applied Biosystems by Life Technologies) ([Supplementary material](#)).

Statistical analysis

SPSS (ver. 21.0, SPSS Inc., Chicago, IL, USA) and R software (ver. 3.1.0, the R Project for Statistical Computing) were used for the statistical analysis. Comparisons between the two groups were performed with Student’s *t*-test or Pearson’s Chi-squared test for quantitative or categorical variables, respectively. Differential abundance levels of genes, taxa, Kyoto Encyclopedia of Genes and Genomes (KEGG) orthologues and gene ontology (GO) terms in the Parkinson’s disease patients and healthy controls were tested using the Wilcoxon rank sum test, and *P*-values were corrected for multiple testing with the Benjamini-Hochberg method for the false discovery rate (FDR) (Storey and Tibshirani, 2003).

Data availability

The high-throughput metagenomic sequence dataset has been deposited in the NCBI BioProject database under project number PRJNA433459. The codes of the key analysis generated in this study have also been uploaded to GitHub (<https://github.com/dryiweiqian/Metagenomics-genes-as-biomarker-of-PD.git>). Other data are available from the corresponding author upon reasonable request.

Results

Diversity and enterotypes of genes in Parkinson’s disease patients and their healthy spouses

The demographic and clinical characteristics of the individuals who participated in this study are summarized in [Tables 1](#) and [2](#). The patients with Parkinson’s disease and healthy control subjects were matched in terms of general demographics, with the exception of constipation and laxative use. Based on the profile of 1 118 355 genes, the within-sample (alpha) diversity (Shannon index) and the between-sample (beta) diversity were used to estimate the gut microbial gene richness and composition of the Parkinson’s disease patients and healthy controls. The alpha diversity of the Parkinson’s disease patients was much higher than that of the healthy controls ($P = 8.38 \times 10^{-3}$, Wilcoxon rank sum test, [Fig. 1A](#)). In addition, a significant difference in the beta diversity between the Parkinson’s disease patients and healthy control subjects was detected [analysis of similarities (ANOSIM) of the Bray-Curtis distance metric, $R = 5.61 \times 10^{-2}$, $P = 9.99 \times 10^{-3}$] ([Fig. 1B](#)). Two distinct enterotypes were clustered from a total of 80 samples ([Fig. 1C](#)). These two enterotypes were primarily composed of several highly abundant genera. *Bacteroides* was the most enriched genus in enterotype 1, whereas *Prevotella* was the most enriched genus in enterotype 2 ([Fig. 1D](#)), as reported in previous studies (Qin *et al.*, 2012; Li *et al.*, 2017a). However, no significant difference in the enterotypes was found between the Parkinson’s disease patients and healthy control subjects ($P = 1$, Fisher’s exact test).

PERMANOVA was performed to analyse the potential effects of different variables on the composition of the gut microbiota. Among the 80 samples, gender, age, BMI, lifestyle factors (cigarette smoking and alcohol, tea, coffee and yoghurt consumption), constipation (along with laxative use) and the use of other medications (proton pump inhibitors, acetylsalicylic acid and statins) had no effect on the microbiome composition. Only Parkinson’s disease status and the enterotype influenced the microbiome composition of the 80 samples ([Supplementary Table 1](#)). Among all the Parkinson’s disease clinical variables present in the 40 patients with Parkinson’s disease, no variable exerted a significant effect on the gut microbiota community ([Supplementary Table 2](#)).

Microbiota profiles of Parkinson’s disease patients and healthy control subjects

As obtained from the metagenomic phylogenetic analysis (MetaPhlAn), the five most abundant phyla and 30 most abundant genera and species in both Parkinson’s disease patients and healthy controls are shown in [Supplementary](#)

Table 1 Demographic characteristics of the subjects in the study

Demographic characteristics	Cohort used for metagenomic sequencing			Cohort used for real-time PCR								
	PD	Healthy	P	PD	Healthy	MSA	MSA-P	AD	P*	P**	P#	P##
n	40	40	–	78	75	40	23	25	–	–	–	–
Age, years	66.6 ± 7.1	66.3 ± 8.1	0.861 ^a	67.0 ± 5.6	65.3 ± 7.6	61.0 ± 6.7	62.5 ± 5.5	66.1 ± 5.2	0.114 ^a	0.000 ^a	0.023 ^a	0.471 ^a
Female, n (%)	21 (52.5)	19 (47.5)	0.848 ^b	37 (47.4)	36 (48.0)	17 (42.5)	11 (47.8)	13 (52.0)	1.000 ^b	0.862 ^b	1.000 ^b	0.843 ^b
BMI, kg/m ² *	23.0 ± 2.6	22.8 ± 2.6	0.801 ^a	23.1 ± 2.8	23.0 ± 2.6	23.9 ± 3.3	24.1 ± 3.1	23.0 ± 3.6	0.824 ^a	0.171 ^a	0.173 ^a	0.869 ^a
Cigarette, n (%)	6 (15.0)	9 (22.5)	0.578 ^b	11 (14.1)	16 (23.3)	4 (10.0)	1 (4.3)	2 (8.0)	0.405 ^b	0.773 ^b	0.456 ^b	0.730 ^b
Alcohol, n (%)	6 (15.0)	1 (2.5)	0.114 ^b	12 (15.4)	15 (20.0)	7 (17.5)	2 (8.7)	4 (16.0)	0.677 ^b	0.799 ^b	0.471 ^b	1.000 ^b
Tea, n (%)	20 (50.0)	20 (50.0)	1.000 ^b	32 (41.0)	34 (45.3)	7 (17.5)	3 (13.0)	3 (12.0)	0.770 ^b	0.071 ^b	0.082 ^b	0.053 ^b
Coffee, n (%)	2 (5.0)	4 (10.0)	0.677 ^b	4 (5.1)	5 (6.7)	5 (7.5)	3 (13.0)	1 (4.0)	0.744 ^b	0.278 ^b	0.355 ^b	1.000 ^b
Yogurt, n (%)	20 (50.0)	16 (40.0)	0.689 ^b	35 (44.9)	28 (37.3)	10 (25.0)	5 (21.7)	1 (4.0)	0.553 ^b	0.185 ^b	0.242 ^b	0.003 ^b
Constipation, n (%)	23 (57.5)	0 (0.0)	0.000 ^b	35 (44.9)	5 (6.7)	27 (67.5)	16 (69.6)	2 (8.0)	0.000 ^b	0.256 ^b	0.325 ^b	0.014 ^b
Laxative use, n (%)	10 (25.0)	0 (0.0)	0.002 ^b	16 (20.5)	3 (4.0)	14 (35.0)	10 (43.5)	0 (0.0)	0.007 ^b	0.209 ^b	0.133 ^b	0.022 ^b
Proton pump inhibitor, n (%)	2 (5.0)	0 (0.0)	0.494 ^b	5 (6.4)	0 (0.0)	2 (5.0)	0 (0.0)	0 (0.0)	0.060 ^b	1.000 ^b	0.583 ^b	0.588 ^b
Acetylsalicylic acid, n (%)	3 (7.5)	2 (5.0)	1.000 ^b	9 (11.6)	8 (10.7)	5 (7.5)	3 (13.0)	1 (4.0)	1.000 ^b	1.000 ^b	1.000 ^b	0.449 ^b
Statin, n (%)	2 (5.0)	0 (0.0)	0.494 ^b	5 (6.4)	0 (0.0)	3 (7.5)	2 (8.7)	0 (0.0)	0.060 ^b	1.000 ^b	0.662 ^b	0.588 ^b
Acetylcholinesterase inhibitor, n (%)	0 (0.0)	0 (0.0)	–	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	12 (48.0)	–	–	–	0.000 ^b
Memantine, n (%)	0 (0.0)	0 (0.0)	–	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	2 (8.0)	–	–	–	0.064 ^b

Differences between two groups were assessed using Student's t-test^a or Fisher's exact test^b. P-values where differences in real-time PCR data between *patients with Parkinson's disease and healthy controls; **patients with Parkinson's disease and patients with MSA; #patients with Parkinson's disease and patients with MSA-P; and ##patients with Parkinson's disease and patients with Alzheimer's disease were detected. UPDRS scores and Hoehn and Yahr stages were obtained from patients during the ON phase at the outpatient clinic. AD = Alzheimer's disease; BMI = body mass index; PD = Parkinson's disease.

Fig. 2. An analysis of the two groups revealed 36 different taxa that were enriched in the Parkinson's disease patients, and no taxon was enriched in the healthy controls (at a prevalence $\geq 10\%$, FDR-corrected $P < 0.05$, Supplementary Table 3). The abundance levels of the kingdoms Viruses and Archaea and of the phyla Synergistetes, Verrucomicrobia and Viruses with no name were higher in the Parkinson's disease patients (Fig. 2A and B). Seven genera differed between the two groups (Fig. 2C). The detection of bacterial species is one of the greatest advantages of shotgun sequencing over 16S rRNA gene sequencing (Ranjan et al., 2016). Among the 13 species enriched in Parkinson's disease patients, 10 belonged to the phylum Firmicutes (Fig. 2D). Among the 40 Parkinson's disease patients, *Streptococcus salivarius* was negatively correlated with LED ($r = -0.4526$, $P = 0.0034$, Supplementary Fig. 3A). *Enterobacter cloacae* was positively correlated with the UPDRS total score ($r = 0.3391$, $P = 0.0347$, Supplementary Fig. 3B). No species was associated with non-motor symptoms (data not shown).

Metagenomic species identification

The 174 964 significantly differentially expressed genes between the Parkinson's disease patients and healthy control subjects were clustered into 153 MGS using a threshold minimum gene number of 50 to explore the microbial genes associated with Parkinson's disease (Supplementary Table 4)

(Segata et al., 2012). After taxonomic characterization, 22 species and 40 genera were identified, with an average genome coverage of 94.1%. Differential abundance levels of MGS and taxonomic characterization of the gut microbiota in Parkinson's disease patients and healthy controls are shown in Fig. 3A. Almost all the identified MGS that were taxonomically enriched in the healthy controls belonged to the genus *Bacteroides*. Consistent with the MetaPhlAn results, the genus *Alistipes* was the major enriched component in Parkinson's disease patients; in particular, the strains *Alistipes indistinctus* YIT 12060 and *Alistipes finegoldii* DSM 17242 were the major components, containing >200 genes. Two MGS annotated to the genus *Akkermansia* and one MGS annotated to the strain *Akkermansia muciniphila* ATCC BAA-835 were enriched in the patients with Parkinson's disease (Fig. 3A and Supplementary Table 6). Importantly, the taxonomic characterizations of most MGS in Parkinson's disease patients are unknown and may be of interest in future studies. A co-occurrence network was generated to investigate potential relationships based on the 31 MGS enriched in the Parkinson's disease patients and the 10 MGS enriched in the healthy controls (Fig. 3B).

Identification of gut microbiota gene markers

The pattern recognition technique based on gut microbiota gene information, as described by Qin et al. (2012, 2014),

Table 2 Clinical characteristics of the subjects in the study

Clinical characteristics	Cohort used for metagenomic sequencing			Cohort used for real-time PCR								
	PD	Healthy	P	PD	Healthy	MSA	MSA-P	AD	P*	P**	P#	P###
Hoehn and Yahr stage	2.3 ± 0.8	–	–	2.3 ± 0.8	–	3.5 ± 1.0	3.6 ± 1.0	–	–	0.000 ^a	0.000 ^a	–
Age of onset, years	60.5 ± 7.5	–	–	60.2 ± 8.3	–	57.1 ± 6.2	58.0 ± 5.3	63.6 ± 5.7	–	0.030 ^a	0.056 ^a	0.059 ^a
Disease duration, years	6.7 ± 4.6	–	–	7.1 ± 4.7	–	3.9 ± 1.9	4.5 ± 1.7	2.6 ± 2.5	–	0.001 ^a	0.002 ^a	0.000h ^a
Motor subtype (tremor), n (%)	21 (52.5)	–	–	37 (47.4)	–	–	–	–	–	–	–	–
Motor subtype (non-tremor), n (%)	19 (47.5)	–	–	41 (52.6)	–	–	–	–	–	–	–	–
UPDRS I score	3.7 ± 2.0	–	–	3.6 ± 2.0	–	–	–	–	–	–	–	–
UPDRS II score	11.7 ± 6.2	–	–	12.4 ± 6.8	–	–	–	–	–	–	–	–
UPDRS III score	26.1 ± 12.3	–	–	27.7 ± 14.1	–	–	–	–	–	–	–	–
UPDRS IV score	3.0 ± 2.7	–	–	2.9 ± 2.7	–	–	–	–	–	–	–	–
UPDRS total score	44.4 ± 19.0	–	–	46.6 ± 21.0	–	–	–	–	–	–	–	–
UMSARS I score	–	–	–	–	–	19.7 ± 6.9	21.1 ± 7.6	–	–	–	–	–
UMSARS II score	–	–	–	–	–	21.3 ± 9.4	25.9 ± 8.7	–	–	–	–	–
UMSARS III score	–	–	–	–	–	0.3 ± 0.4	0.2 ± 0.4	–	–	–	–	–
UMSARS IV score	–	–	–	–	–	2.2 ± 1.1	2.6 ± 1.0	–	–	–	–	–
UMSARS total score	–	–	–	–	–	43.5 ± 16.0	49.8 ± 15.7	–	–	–	–	–
NMS score	7.9 ± 4.2	–	–	8.2 ± 4.3	–	10.5 ± 2.5	10.3 ± 2.1	–	–	0.001 ^a	0.029 ^a	–
HAMD score	6.8 ± 7.0	–	–	6.6 ± 6.9	–	11.6 ± 4.7	12.4 ± 5.1	–	–	0.000 ^a	0.000 ^a	–
HAMA score	9.2 ± 6.7	–	–	9.1 ± 6.7	–	12.2 ± 5.1	12.5 ± 4.8	–	–	0.007 ^a	0.010 ^a	–
MMSE score	28.1 ± 2.6	–	–	27.7 ± 3.1	–	26.0 ± 3.1	25.6 ± 3.3	19.0 ± 5.6	–	0.008 ^a	0.010 ^a	0.000 ^a
PD medication (n, %)	40 (100.0)	–	–	71 (91.0)	–	39 (97.5)	23 (100.0)	–	–	0.889 ^b	0.866 ^b	–
Levodopa	35 (87.5)	–	–	68 (87.2)	–	32 (80.0)	23 (100.0)	–	–	0.775 ^b	0.736 ^b	–
Dopamine agonists	25 (62.5)	–	–	48 (61.5)	–	16 (40.0)	13 (56.5)	–	–	0.242 ^b	1.000 ^b	–
COMT inhibitor	2 (5.0)	–	–	5 (6.4)	–	5 (12.5)	5 (21.7)	–	–	0.320 ^b	0.118 ^b	–
MAOB inhibitor	10 (25.0)	–	–	18 (23.1)	–	9 (22.5)	5 (21.7)	–	–	1.000 ^b	1.000 ^b	–
Benzhexol hydrochloride	5 (12.5)	–	–	9 (11.5)	–	1 (2.5)	1 (4.3)	–	–	0.167 ^b	0.687 ^b	–
Amantadine	4 (10.0)	–	–	9 (11.5)	–	6 (15.0)	5 (21.7)	–	–	0.774 ^b	0.324 ^b	–
LED mg/day	428.4 ± 257.8	–	–	434.8 ± 260.1	–	563.8 ± 405.6	802.2 ± 313.6	–	–	0.039 ^a	0.000 ^a	–
Motor complications, n (%)	18 (45.0)	–	–	45 (57.7)	–	–	–	–	–	–	–	–

Differences between two groups were assessed using Student's *t*-test^a or Fisher's exact test^b. *P*-values where differences in real-time PCR data between *patients with Parkinson's disease and healthy controls; **patients with Parkinson's disease and patients with MSA; #patients with Parkinson's disease and patients with MSA-P; and ###patients with Parkinson's disease and patients with Alzheimer's disease were detected.

UPDRS scores and Hoehn and Yahr stages were obtained from patients during the ON phase at the outpatient clinic. AD = Alzheimer's disease; HAMA = Hamilton Anxiety Scale; HAMD = Hamilton Depression Scale; LED = levodopa equivalent dose; MMSE = Mini-Mental State Examination; NMS = non-motor symptom; PD = Parkinson's disease; UMSARS = Unified MSA Rating Scale; UPDRS = Unified Parkinson's Disease Rating Scale.

was used to identify patients with Parkinson's disease in our study. Twenty-five optimal gene markers were obtained from 51 816 genes using the mRMR method (Supplementary Fig. 4). An SVM discriminator was also constructed, and the AUC was 0.896 (95% CI: 83.1–96.1%), with a sensitivity of 0.90 and a specificity of 0.75 (Fig. 4A). A more straightforward index for patient discrimination, the PDI, was defined to facilitate the clinical application of the selected microbial gene markers. The average PDI was significantly different between the Parkinson's disease patients and the healthy control subjects (Fig. 4A, $P = 2.458 \times 10^{-11}$, Wilcoxon rank sum test).

A PERMANOVA was conducted to examine potential confounders of the PDI and further confirm the possible effects of both the demographic and Parkinson's disease clinical characteristics on the identified PDI based on the 25 gene markers. Among the 80 samples, only Parkinson's disease status influenced the PDI ($P = 0.0003$, Supplementary Table 5). More importantly, in the 40 patients with

Parkinson's disease, the identified PDI was not influenced by disease severity or the use of Parkinson's disease medications (Supplementary Table 6), suggesting that the PDI represented a potentially stable diagnostic biomarker of Parkinson's disease.

Application of genetic markers for identifying Parkinson's disease patients using real-time PCR

We constructed plasmids carrying the 25 genes used as the standards in the real-time PCR analysis to increase the convenience of disease index testing (Fig. 4B). Using the real-time PCR method to test the 25 gene markers, the AUC value obtained for the Parkinson's disease patients and healthy controls for whom shotgun sequencing was performed was 0.922 (95% CI: 86.4–98.0%), with a sensitivity of 0.95 and a specificity of 0.75 (Fig. 4C). The PDIs of the

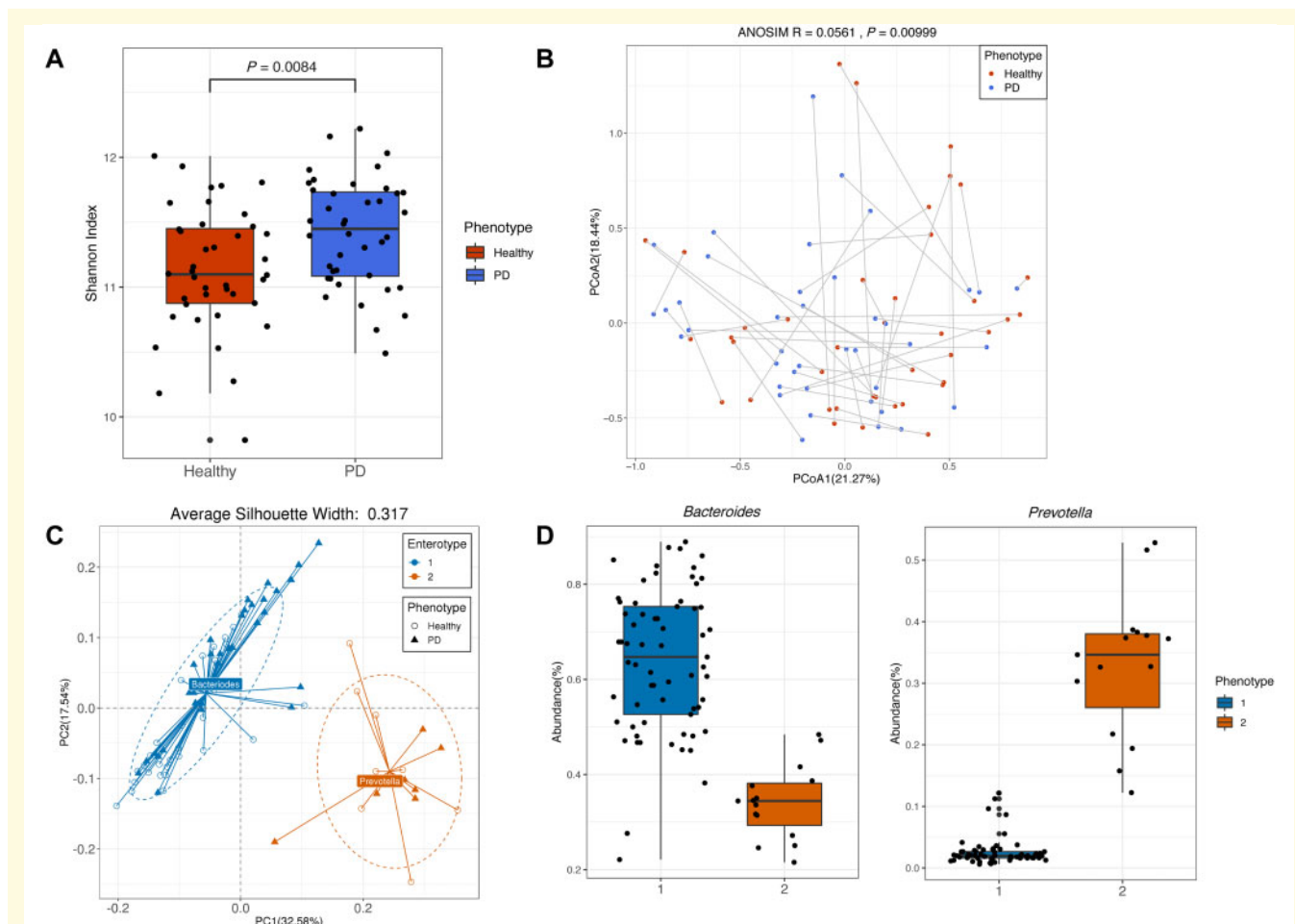


Figure 1 Differences in the diversity and enterotypes of gut microbial genes between patients with Parkinson's disease and healthy control subjects. **(A)** The difference in the Shannon index (alpha diversity) between the Parkinson's disease patients and healthy control subjects was based on 1118355 gut microbial genes. Blue and red indicate Parkinson's disease patients and healthy controls, respectively. **(B)** Unweighted ANOSIM was performed with the Bray-Curtis distance matrix (beta diversity) of the gut microbial compositions in the Parkinson's disease patients and healthy controls based on the 1118355 gut microbial genes. The ANOSIM R-value indicated the community variation between the two groups, and a significant P-value was obtained. The two dimensions explained the greatest proportion of variance in the communities. Each symbol represents a sample, and each line connects a pair of samples. **(C)** Overall, 80 samples were clustered into enterotype 1 (blue) and enterotype 2 (orange) based on the abundance levels of genera determined using the principal component analysis of Jensen-Shannon distance. The average silhouette width was used to determine the optimal number of clusters. **(D)** The abundance levels of the main contributing genera of each enterotype are shown. Each box represented the interquartile range (IQR), and the lines in the boxes indicated the median values. The whiskers showed the lowest and highest values within the IQR from the first and third quartiles. All data-points are shown. PD = Parkinson's disease.

Parkinson's disease patients were significantly different from those of the healthy controls (Fig. 4C, $P = 5.014 \times 10^{-13}$, Wilcoxon rank sum test). The AUC, sensitivity and specificity results obtained from real-time PCR testing of the 25 gene markers were in good accordance with the results from metagenomic sequencing.

We then validated the discriminatory power of the 25 gene markers using real-time PCR in another larger independent group including 78 patients with Parkinson's disease and 75 healthy control subjects and obtained an AUC of 0.905 (95% CI: 86.0–95.1%, sensitivity = 0.86, specificity = 0.77) (Fig. 4D), suggesting that the PDI classifier can be applied to discriminate Parkinson's disease patients from

healthy controls (Fig. 4D, $P = 1.341 \times 10^{-21}$, Wilcoxon rank sum test). A cohort of 40 MSA patients (23 MSA-P patients and 17 MSA-C patients) was enrolled to investigate the disease specificity of these 25 gene markers. An AUC of 0.831 (95% CI: 74.0–92.2%, sensitivity = 0.85, specificity = 0.78) was obtained to distinguish between the 78 patients with Parkinson's disease and 40 MSA patients for whom different PDI values were obtained (Fig. 4E, $P = 5.482 \times 10^{-10}$, Wilcoxon rank sum test). Interestingly, the PDI values were significantly different between the 78 Parkinson's disease patients and the 23 MSA-P patients ($P = 9.200 \times 10^{-6}$, Wilcoxon rank sum test), with an AUC of 0.793 (Fig. 4F, 95% CI: 67.5–91.1%, sensitivity = 0.88, specificity = 0.70).



Additionally, an AUC of 0.901 (95% CI: 82.7–97.6%, sensitivity = 0.90, specificity = 0.88) was obtained for distinguishing between the 78 Parkinson’s disease patients and a cohort of 25 Alzheimer’s disease patients, and significantly different PDI values were also found (Fig. 4G, $P = 2.548 \times 10^{-11}$, Wilcoxon rank sum test). All the actual subject numbers among the different cohorts are provided in the confusion matrix table (Supplementary Table 7).

The most abundant KEGG orthologue and KEGG enrichment pathways that were annotated were analysed. The most abundant KEGG orthologues (containing >20 genes) were involved in metabolism, including the global and overview maps, amino acid metabolism, carbohydrate metabolism, metabolism of cofactors and vitamins, and energy metabolism (Fig. 5A). The most prevalent pathways involved vancomycin resistance, biotin metabolism, other glycan degradation, and phenylalanine, tyrosine and tryptophan biosynthesis (Supplementary Fig. 5A and Supplementary Table 8).

(Fig. 5B). Among the top 30 enriched GO terms, the main enriched GO terms were related to biological processes (e.g. pyrimidine deoxyribonucleotide catabolism and phyloquinone biosynthesis) and molecular functions (e.g. anthranilate synthase activity) (Supplementary Fig. 5B and Supplementary Table 9).

We annotated the 25 gene markers using the NCBI non-redundant protein sequence database, and 23 gene markers were annotated. Interestingly, we found that 10 markers were annotated to the genus *Bacteroides* and that seven markers were annotated to the species *Bacteroides coprocola*, which were also less common in the Parkinson's disease patients in the MGS identification procedure described above (Fig. 3A). Marker 21 was annotated to the WP_087393524.1 hypothetical protein of *Akkermansia muciniphila* (Supplementary Table 10). Detailed information regarding the 25 gene markers is listed in Supplementary Table 11.

We established the first gut microbial gene catalogue associated with Parkinson's disease on the basis of metagenomic sequencing. More importantly, we constructed a set of 25 gut microbial gene markers to identify patients with Parkinson's disease, representing the first set of its kind. In particular, the PDI based on the 25 gene markers was not influenced by the demographic or Parkinson's disease clinical characteristics, indicating that the PDI represents a potentially valuable biomarker for diagnosing Parkinson's disease. Furthermore, we used a rapid and economic real-time PCR method to test these 25 gene markers with high sensitivity

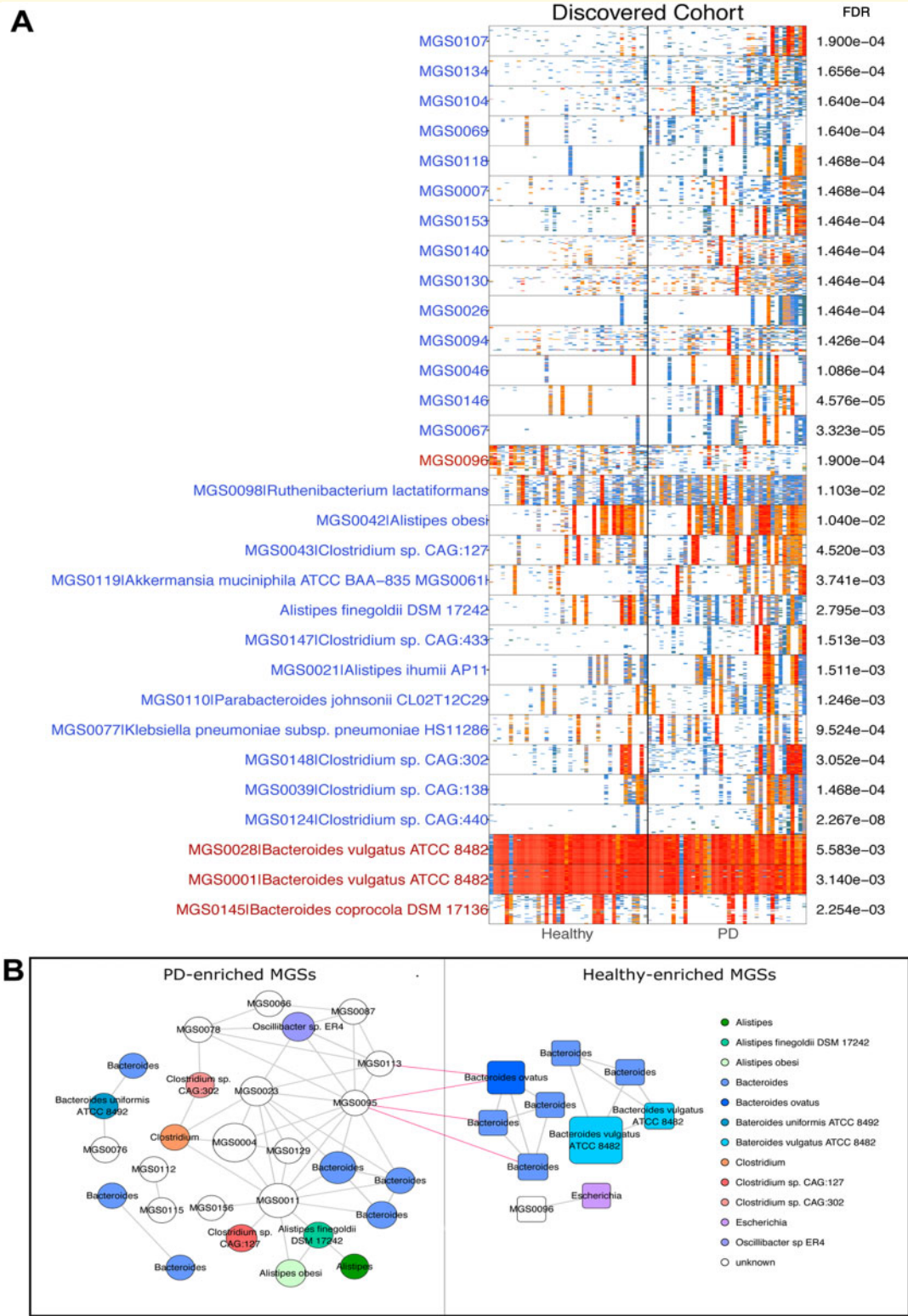


Figure 3 Differential abundance levels of MGS and taxonomic characterization of the gut microbiota in Parkinson's disease patients and healthy controls. (A) The presence and abundance of 30 representative differential MGS containing at least 50 'tracer' genes are shown. Mann-Whitney probabilities (P-value, FDR-adjusted) are given. Genes are in rows, and the frequency is indicated by a colour gradient (white, not detected; red, most abundant). Individual subjects are in columns and ordered by increasing gene number. Blue and red in MGS indicate Parkinson's disease patients and healthy control subjects, respectively. **(B)** A co-occurrence network was constructed from 31 MGS enriched in Parkinson's disease patients and 10 MGS enriched in healthy controls. Nodes indicate MGS, with their IDs displayed in the centre. The node size is proportional to the number of genes contained in the MGS. The colour of a node indicates its taxonomic assignment. Connecting lines indicate Spearman correlation coefficient values >0.7. PD = Parkinson's disease.

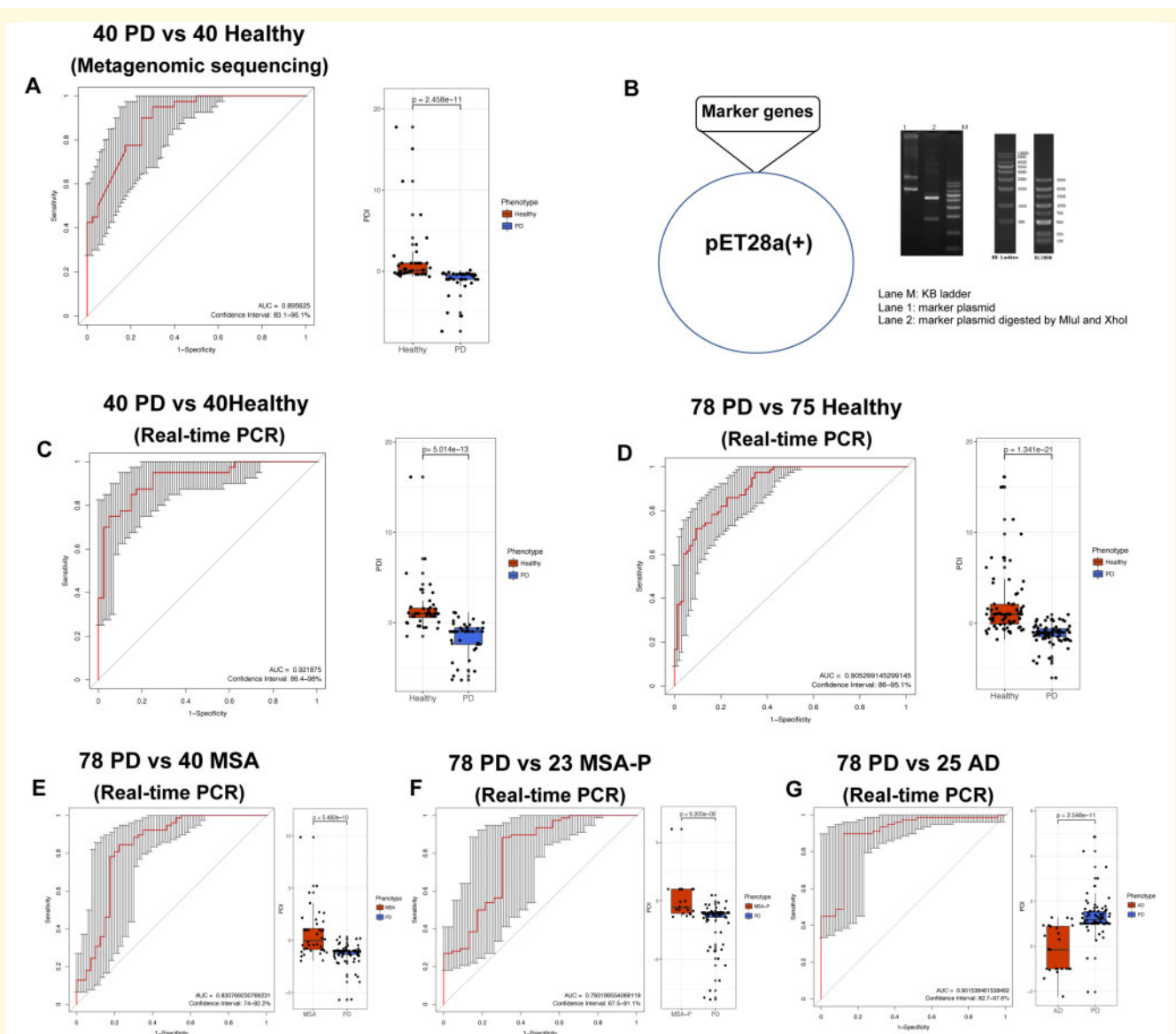


Figure 4 Evaluation of the risk of Parkinson's disease based on 25 gut microbial markers. (A) Twenty-five genes were selected using the mRMR approach from shotgun metagenomic sequencing to evaluate the risk of Parkinson's disease. The AUC and computed PDI are shown for 40 Parkinson's disease patients and 40 healthy control subjects. (B) Plasmid DNAs containing each of the 25 gene marker fragments were synthesized and inserted into pET28a(+) vectors as DNA standards for real-time PCR. (C) The training samples were 40 Parkinson's disease patients and 40 healthy controls from metagenomic sequencing. The AUC and computed PDI based on real-time PCR are shown. (D) The validation samples using real-time PCR included an additional 78 Parkinson's disease patients and 75 healthy control subjects. The AUC and computed PDI are shown. (E) The 25 gene markers were also validated using real-time PCR in a cohort of 40 MSA patients. The AUC and computed PDI are shown for 78 Parkinson's disease patients and 40 MSA patients. (F) The AUC and computed PDI are shown for 78 Parkinson's disease patients and 23 MSA-P patients. (G) The AUC and computed PDI are shown for 78 Parkinson's disease patients and 25 Alzheimer's disease patients. Each box depicts the IQR between the first and third quartiles (25th and 75th percentiles, respectively), and the line inside denotes the median. All data-points are shown. AD = Alzheimer's disease; PD = Parkinson's disease.

and specificity, similar to metagenomic sequencing. Additionally, the identified PDI was consistently validated not only in a larger independent cohort of Parkinson's disease patients but also in MSA patients and patients with other neurodegenerative diseases, i.e. Alzheimer's disease.

Previous proposed diagnostic models of Parkinson's disease using 16S sequencing focused on taxonomic information, but not all taxa included in each study were the same (Keshavarzian *et al.*, 2015; Scheperjans *et al.*, 2015; Hill-Burns *et al.*, 2017; Hopfner *et al.*, 2017; Li *et al.*, 2017b;

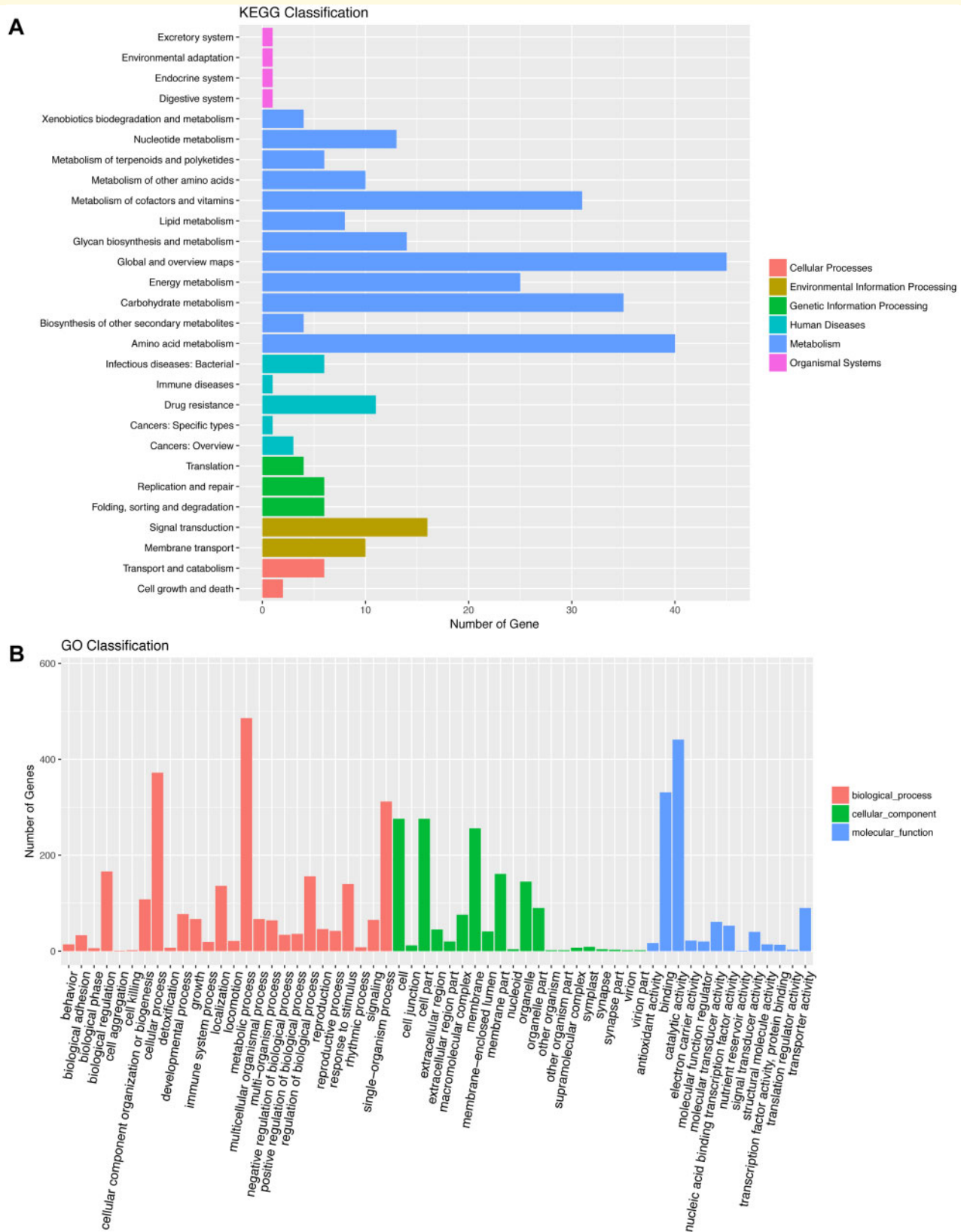


Figure 5 Different abundance levels of genes annotated according to functional classification. The significantly differentially expressed genes between the Parkinson's disease patients and healthy control subjects were annotated according to KEGG (A) and GO classifications (B). The number of gene hits is shown along the y-axis, and the different KEGG (A) and GO (B) classifications are shown along the x-axis. $P_{FDR} < 0.05$ was used as the threshold for selecting significant KEGG and GO classifications.

Petrov *et al.*, 2017; Heintz-Buschart *et al.*, 2018; Lin *et al.*, 2018; Qian *et al.*, 2018). Furthermore, none of these studies enrolled an independent validation cohort to test the diagnostic power of the method. We enrolled Parkinson's disease patients' healthy spouses as controls to mitigate the influence of diet on the microbiota and identify significant changes associated with the disease itself, a strategy that has been used in recent studies (Qian *et al.*, 2018; Tan *et al.*, 2018). Moreover, the diagnostic models of Parkinson's disease that we proposed were based on gene information rather than taxonomy. The PDI based on the 25 identified gene markers was not influenced by dietary habits, constipation, laxative use, Parkinson's disease severity or any Parkinson's disease medications, indicating that the PDI represented a stable diagnostic biomarker of Parkinson's disease. Moreover, the specificity of the identified markers in our study was comparable to those of markers reported in other studies, with a substantially higher AUC (0.896 versus 0.80–0.84) and sensitivity (0.90 versus 0.65–0.76) (Scheperjans *et al.*, 2015; Bedarf *et al.*, 2017; Lin *et al.*, 2018; Qian *et al.*, 2018). Furthermore, we validated these genes in a larger independent cohort to confirm the replicability of these markers.

MWAS is a good method for identifying gut microbiome-based gene markers for the diagnosis of diseases (Wang and Jia, 2016) as 15 markers have been identified for liver cirrhosis (Qin *et al.*, 2014) and 50 markers have been identified for type 2 diabetes (Qin *et al.*, 2012). However, these gene markers proposed by the MWAS were subsequently validated using metagenomic sequencing (Qin *et al.*, 2012, 2014), which is accurate but expensive and therefore has limited applications. Based on the strategy used in the colorectal cancer project (Petrov *et al.*, 2017), which is the only study to verify genetic markers using real-time PCR, we validated the gene markers identified by MWAS using real-time PCR in the original cohort and another larger independent cohort. We obtained similar AUC, specificity and sensitivity values, confirming the consistency of these markers. Moreover, both MSA and Parkinson's disease are movement disorders, and clinicians cannot easily diagnose and distinguish MSA patients, particularly MSA-P patients, from Parkinson's disease patients at the early stage. Considering that some evidence indicates a likelihood that each disease has been misdiagnosed based on the extensive clinicopathological literature, our results provide a tool for clinical diagnosis. Recently, gut microbiota dysbiosis was also reported in patients with MSA (Tan *et al.*, 2018; Du *et al.*, 2019; Wan *et al.*, 2019) and Alzheimer's disease (Vogt *et al.*, 2017; Zhuang *et al.*, 2018; Li *et al.*, 2019a; Liu *et al.*, 2019). Interestingly, the set of 25 gene markers identified in our study was used to distinguish Parkinson's disease patients not only from normal controls but also from MSA patients and Alzheimer's disease patients. The power of the set of 25 gene markers to distinguish between Alzheimer's disease and Parkinson's disease patients was much higher than that between MSA and Parkinson's disease patients. Considering that Alzheimer's disease is one of the most common neurodegenerative diseases without associated

movement problems, this panel of 25 gene sets may be specific to movement disorders. Overall, the gut microbial gene markers identified here have important disease-specific diagnostic value and considerable potential for clinical applications.

In addition to the gene markers, only Parkinson's disease status and the enterotype influenced the microbiome composition. The concept of the 'enterotype' refers to stratification of human gut microbiota. The essence of enterotyping is a process of dimensionality reduction to collapse global microbiome variation into a few categories (Arumugam *et al.*, 2011; Cheng and Ning, 2019). Some studies stratified the microbiome into three enterotypes driven by discriminative genera, including *Bacteroides* (enterotype 1), *Prevotella* (enterotype 2), and *Ruminococcus* (enterotype 3) (Arumugam *et al.*, 2011; Qin *et al.*, 2012). Moreover, several studies have confirmed that samples can be stratified into only two enterotypes represented by *Bacteroides* and *Prevotella* (Lim *et al.*, 2014; Zhang *et al.*, 2014; Nakayama *et al.*, 2017). The alteration of microbiota composition in the short term might not be sufficient to switch the enterotype (Cheng and Ning, 2019). Some researchers found that enterotypes could not be affected by gender, weight or geographical factors, showing relatively high stability (Costea *et al.*, 2018). Dietary intake and administration of antibiotics are known to have a possible impact on both enterotype patterns and identification (Haikal *et al.*, 2019). In our study, Parkinson's disease patients' healthy spouses were enrolled as controls to mitigate the influence of diet. Moreover, all included participants did not use antibiotics within 3 months prior to sample collection. Our findings indicated that the main factor affecting the gut microbial composition of Parkinson's disease patients was the disease itself.

We found that the microbiota composition does not appear to be altered by anything other than Parkinson's disease status using the same statistical method as these studies (Bedarf *et al.*, 2017; Hill-Burns *et al.*, 2017). This finding supports the hypothesis that gut microbial genes are potential markers of Parkinson's disease. The observed effects of Parkinson's disease medications on the microbiome were concluded from 16S sequencing (Scheperjans *et al.*, 2015; Hill-Burns *et al.*, 2017). Bedarf *et al.* (2017) detected that non-levodopa medication had no effects on microbiota abundance and function, which is similar to our finding that anti-parkinsonism medication had no influence on microbiota composition (Supplementary Table 12). We also found that constipation did not affect the microbiome, as there were no constipated controls enrolled in our study, which is a limitation. Enrolling control groups with constipation for more research is needed in the future. The lifestyle, gender, age and BMI of the two groups in our study were matched. Diet is an important factor affecting the gut microbiota. All the enrolled couples had lived together for at least 20 years, and the couples shared more similar bacterial communities in their guts compared to unrelated individuals living in other households (Song *et al.*, 2013). In particular, the

dietary style in China (eating together) is different from Western diet habits (individual servings). Spouses of cases were enrolled to minimize the influence of diet, but it is impossible to control for diet in the statistical analysis, which is also a limitation of our study.

Our research also added new information to the taxonomic and functional aspects associated with Parkinson's disease. Based on our metagenomic sequencing results, the diversity and community of gut microbial genes in Parkinson's disease patients differed from those of healthy control subjects, indicating that gut microbiota dysbiosis exists in Parkinson's disease patients, which is consistent with findings from other studies (Keshavarzian et al., 2015; Scheperjans et al., 2015; Hill-Burns et al., 2017; Hopfner et al., 2017; Petrov et al., 2017). Interestingly, seven markers that we identified were annotated to the species *Bacteroides coprocola*, with a lower abundance in Parkinson's disease patients, which has also been reported by Petrov et al. (2017). The gut microbiome can be influenced by race and nationality. Based on related studies, consistent key bacteria associated with Parkinson's disease could not be obtained from a series of studies conducted around the world (Hasegawa et al., 2015; Keshavarzian et al., 2015; Scheperjans et al., 2015; Unger et al., 2016; Bedarf et al., 2017; Hill-Burns et al., 2017; Hopfner et al., 2017; Li et al., 2017b, 2019b; Minato et al., 2017; Petrov et al., 2017; Heintz-Buschart et al., 2018; Lin et al., 2018, 2019; Qian et al., 2018; Aho et al., 2019; Barichella et al., 2019; Haikal et al., 2019; Pietrucci et al., 2019). The genera *Akkermansia* and *Prevotella* were the bacteria that garnered the most attention. The genus *Akkermansia* (Keshavarzian et al., 2015; Unger et al., 2016; Bedarf et al., 2017; Hill-Burns et al., 2017; Heintz-Buschart et al., 2018; Barichella et al., 2019; Li et al., 2019b; Lin et al., 2019) and the species *Akkermansia muciniphila* (Bedarf et al., 2017; Heintz-Buschart et al., 2018) were consistently found to be more abundant in Parkinson's disease patients. As obtained from the MetaPhlAn, *Akkermansia*/*Akkermansia muciniphila* were not significantly different between the Parkinson's disease and control groups, but the abundance of *Akkermansia*/*Akkermansia muciniphila* tended to increase in Parkinson's disease ($P_{\text{FDR}} < 0.07$, Supplementary Fig. 6). Furthermore, from MGS identification, we found two MGS annotated to the genus *Akkermansia* and one MGS annotated to the strain *Akkermansia muciniphila* ATCC BAA-835, which were all enriched in Parkinson's disease patients. More importantly, marker 21 was annotated to the WP_087393524.1 protein of *Akkermansia muciniphila*, suggesting that this *Akkermansia*-related protein may play an important role in Parkinson's disease progression. Consistent with our previous 16S study, the reduction in *Prevotellaceae*/*Prevotella* was also not found in the Parkinson's disease patients in this metagenomics study (Qian et al., 2018). The decreased abundance of *Prevotellaceae*/*Prevotella* in Parkinson's disease cases was found in several studies that were mainly obtained from German and Russian populations (Scheperjans et al., 2015; Unger et al., 2016; Bedarf et al.,

2017; Hopfner et al., 2017; Petrov et al., 2017), since it was not consistent with studies from the Asian population (Hasegawa et al., 2015; Li et al., 2017b, 2019b; Lin et al., 2018; Qian et al., 2018) [except one study from Taiwan (Lin et al., 2019)]. The difference may be affected by race and nationality. Given our results, we proposed that gut microbial genes may be a more stable biomarker than bacteria. At present, only metagenomic-based markers were proposed in our study. However, as the identification and validation of these gene markers were based on subjects with a Han Chinese ethnic background, there can be limitations to applying the results to other populations. Further validation of these gut microbial gene markers in larger cohorts across different populations is needed.

We identified the kingdom Viruses (phylum Viruses no name/unclassified) as enriched in Parkinson's disease patients. This finding conflicts with those in studies by Bedarf and colleagues (2017), and the difference may be related to the limited number of reference genomes in the ACLAME database used in the previous study. Viruses, such as influenza virus, Coxsackie virus, Japanese encephalitis virus, and HIV virus, can induce secondary Parkinson's disease (Jang et al., 2009). Recently, a study from Taiwan showed that hepatitis C virus infection is associated with the risk of Parkinson's disease (Tsai et al., 2016). We were unable to specify the types of viruses related to Parkinson's disease in the present study; the occurrence of viruses as well as bacteria warrants future exploration.

According to emerging evidence, gut microbial metabolism has direct impact on human health (Puertollano et al., 2014). We found that the gut microbiome mostly affected metabolism-related functions in Parkinson's disease patients, e.g. amino acid metabolism and phenylalanine, tyrosine, and tryptophan biosynthesis, which is consistent with the results of Bedarf et al. (2017). Alterations of metabolites such as catecholamines, serotonin, amino acids [including glutamate, homocysteine and large neutral amino acids (LNNA)-tyrosine, phenylalanine, tryptophan or branched-chain amino acids (BCAA)] in serum/plasma, CSF or urine of Parkinson's disease patients have been reported (Havelund et al., 2017). Both the clinical (Luan et al., 2015; LeWitt et al., 2017) and animal studies (Lu et al., 2014; Shukla et al., 2016) reported these changes in alanine metabolism, BCAA and fatty acid metabolism, leading to mitochondrial dysfunction, which is the underlying pathophysiological and pathogenetic mechanism in Parkinson's disease. We also confirmed that cofactor and vitamin metabolism and energy metabolism were associated with Parkinson's disease, which we predicted using 16S sequencing (Qian et al., 2018). The association between vitamin metabolism and Parkinson's disease may explain the vitamin deficiency in Parkinson's disease patients to some extent (Evatt et al., 2008; Shen, 2015).

Collectively considering our results, we established the first collection of Parkinson's disease gut microbiomes in the world. The identified PDI based on the set of 25 gut microbial gene markers that we selected effectively distinguished Parkinson's disease patients from both healthy controls and

patients with MSA or Alzheimer's disease; moreover, the PDI was not influenced by disease severity or the use of Parkinson's disease medications. These gut microbial gene markers were tested using a real-time PCR method with the same efficiency as metagenomic sequencing. Based on our results, the identified PDI may be a potentially useful diagnostic biomarker of Parkinson's disease.

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Competing interests

The authors report no competing interests.

Supplementary material

[Supplementary material](#) is available at *Brain* online.

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